

Algebra: Piecewise Defined Functions

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Name: _____ Date: _____

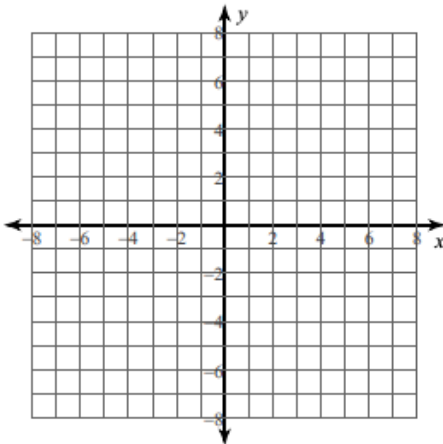
Score: / 8

DIRECTIONS

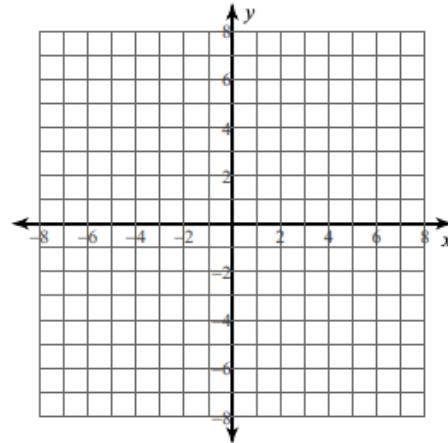
Evaluate the piecewise function at each x -value by choosing the correct piece.

Sketch the graph of each function.

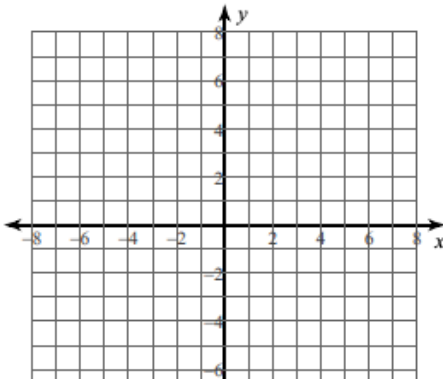
1) $f(x) = \begin{cases} -2x - 1, & x \leq 2 \\ -x + 4, & x > 2 \end{cases}$ $f(3) = \boxed{}$



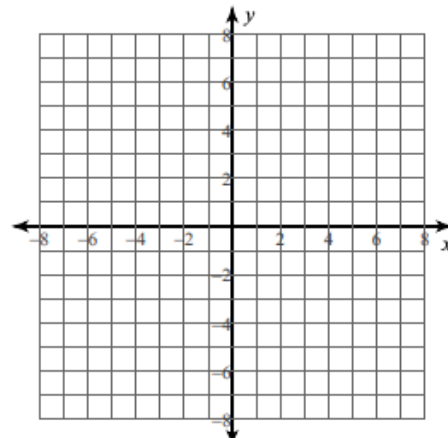
2) $f(x) = \begin{cases} -4, & x \leq -2 \\ x - 2, & -2 < x < 2 \\ -2x + 4, & x \geq 2 \end{cases}$ $f(2) = \boxed{}$



3) $f(x) = \begin{cases} -2^x, & x < -4 \\ -|x|, & -4 \leq x \leq 0 \\ 4 - x^2, & x > 0 \end{cases}$ $f(-2) = \boxed{}$



4) $g(x) = \begin{cases} -6, & x < -2 \\ (x + 1)^4, & x \geq -2 \end{cases}$ $f(-2) = \boxed{}$



Answer Key & Solutions

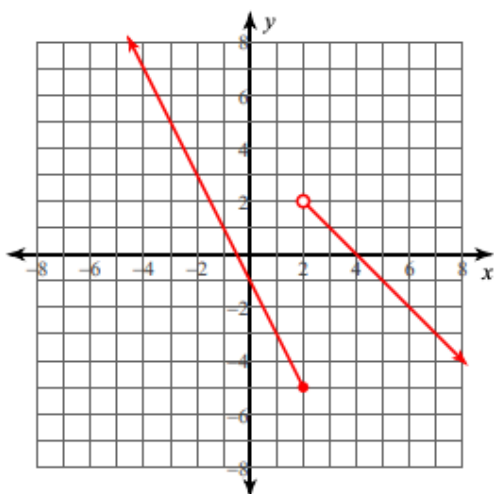
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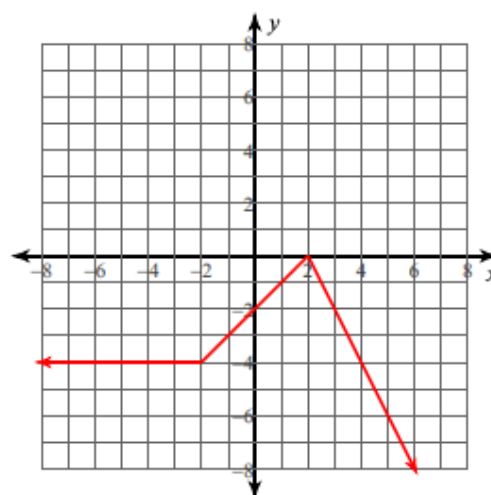
TEACHER NOTES Students must check the condition for x first, then substitute into the matching rule.

Sketch the graph of each function.

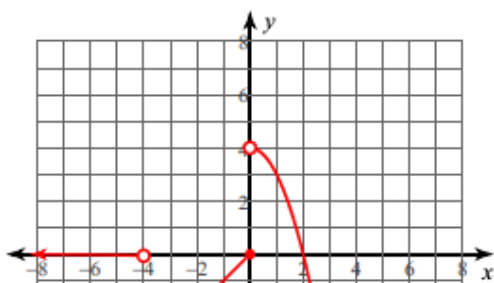
1) $f(x) = \begin{cases} -2x - 1, & x \leq 2 \\ -x + 4, & x > 2 \end{cases}$ $f(3) =$ 1



2) $f(x) = \begin{cases} -4, & x \leq -2 \\ x - 2, & -2 < x < 2 \\ -2x + 4, & x \geq 2 \end{cases}$ $f(2) =$ 0



3) $f(x) = \begin{cases} -2^x, & x < -4 \\ -|x|, & -4 \leq x \leq 0 \\ 4 - x^2, & x > 0 \end{cases}$ $f(-2) =$ -2



4) $g(x) = \begin{cases} -6, & x < -2 \\ (x + 1)^4, & x \geq -2 \end{cases}$ $f(-2) =$ 1

